Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Project Documentation

1. Introduction

- **Project Title:** Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
- **Team Members:**
  - Team Member 1: Project lead
  - Team Member 2: ServiceNow developer (Flow Designer and catalog)
  - Team Member 3: Business analyst (requirements and testing)
  - Team Member 4: IT procurement coordinator (process owner and UAT)
  - Team Member 5: Tester (test cases, bug tracking and documentation)

2. Project Overview

- **Purpose:** Automate the ordering of standard laptops in ServiceNow so employees get approved devices faster and IT spends less time on repetitive procurement work.
- **Features:**
  - Standard laptop catalog item with details pre-filled from the user record and validation of eligibility and duplicates
  - Automatic approval for pre-approved roles, or one-click manager approval with escalation after 2 days
  - Automatic stock check and reservation, with purchase order creation when stock is low
  - Asset record creation, stage-by-stage notifications and a turnaround and savings dashboard

3. Architecture

- **Presentation layer (in place of Frontend):** Employee Center / Service Portal for requests and status, plus email and mobile approvals for managers.
- **Application layer (in place of Backend):** ServiceNow Service Catalog and Flow Designer. One main flow calls three subflows: approval, stock check and purchase order, and asset record and notification.
- **Data layer (in place of Database):** ServiceNow tables: sc_request and sc_req_item for requests, alm_hardware for asset inventory and proc_po for purchase orders. Flows read and write these tables through Flow Designer actions.

4. Setup Instructions

- **Prerequisites:**
  - A ServiceNow instance (for example, a Personal Developer Instance)
  - Admin access, and test users for the employee, manager and coordinator roles
  - Email notifications enabled on the instance, and access to the vendor endpoint for purchase orders
- **Installation:**
  - 1. Request or open your ServiceNow instance and log in as an administrator.
  - 2. Create the Standard Laptop catalog item and its variables (role, location, department, cost center).
  - 3. Create the user roles and assign the test users.
  - 4. Build the main flow and the three subflows in Flow Designer, then activate them.
  - 5. Configure the notifications and the vendor REST message with its credentials.
  - 6. Load sample laptop models into the asset inventory and set stock levels.

5. Folder Structure

ServiceNow stores work as records inside an application scope instead of folders, so the structure below shows how the components are organized.

Component	Contents
Catalog (in place of Client)	Standard Laptop catalog item, variables and the request form
Flows (in place of Server)	Main laptop order flow and subflows: approval, stock check and purchase order, asset record and notification
Actions and rules	Custom actions, decision rules for pre-approved roles and the escalation timer
Notifications	Email templates for approved, shipped and delivered
Reports	Dashboard and reports for turnaround time, status and savings

6. Running the Application

- **Employee view:** open Employee Center, select Standard Laptop and submit the request.
- **Flow check:** open Flow Designer, then Executions, to follow the request through each step.
- **Manager view:** approve the task from the email or the approvals list.
- **Coordinator view:** check stock, purchase orders and asset records.

7. API Documentation

The solution uses ServiceNow's Table API for reading records and one outbound REST message to send purchase orders to the vendor. The endpoints below are examples.

Method	Endpoint	Parameters	Description
GET	`/api/now/table/sc_request`	sysparm_query, sysparm_limit	Lists laptop requests, for example filtered by state
GET	`/api/now/table/alm_hardware`	sysparm_query	Checks stock for a laptop model
POST	`[vendor endpoint]/purchase-orders`	Model, quantity, ship-to location	Sends a purchase order to the vendor (outbound REST message)

Example response for the vendor purchase order call:

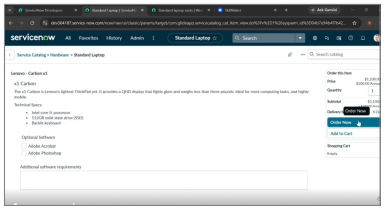
```
{
  "po_number": "PO0001234",
  "status": "accepted",
  "expected_delivery": "2025-03-05"
}
```

8. Authentication

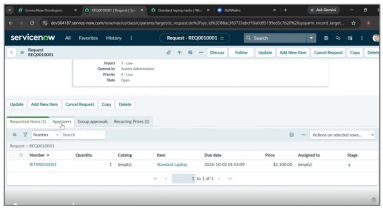
- Users sign in through the identity provider using single sign-on (SAML).
- Authorization uses ServiceNow roles and access control rules, so only managers can approve and only authorized users can see request and asset data.
- REST calls to the instance use OAuth or basic authentication with an integration user. The vendor call uses the vendor's API key, stored in the credential record rather than in the flow.

9. User Interface

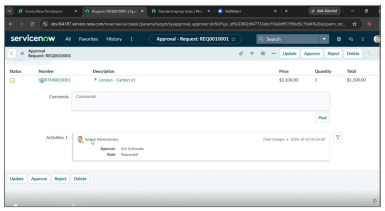
Add screenshots of the main screens below.



Standard laptop request form



Request status page (request record)



Manager approval (ServiceNow approval record)

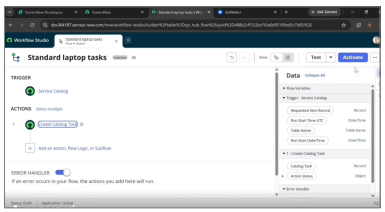
Add screenshot
Turnaround and savings dashboard

10. Testing

- **Strategy:** functional testing of each flow step, performance testing for success rate, execution time and load, and user acceptance testing with the employee, manager and coordinator roles.
- **Tools:** Flow Designer execution details, ServiceNow Automated Test Framework (ATF), a test instance and the UAT template.

11. Screenshots or Demo

Add the demo link and screenshots of the Flow Designer flow and a completed request.



Flow Designer flow

Add screenshot
Completed request with asset record

Demo link: [add demo link here]

12. Known Issues

- The approval button is hard to tap on small mobile screens (BG-003, in progress).
- Vendor delivery times are not tracked yet, so delivery dates come from the vendor response only.

13. Future Enhancements

- Extend the same flow to other standard hardware such as monitors, phones and accessories.
- Compare vendor prices automatically before creating a purchase order.
- Add a mobile-friendly approval screen and chatbot-based requests.
- Forecast stock needs from hiring plans and refresh cycles.

Note: the team roles, endpoints, example response and known issues are sample entries. Replace them with your project's real details before you submit.